Basic Syntax

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Agenda

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4. Basic syntax
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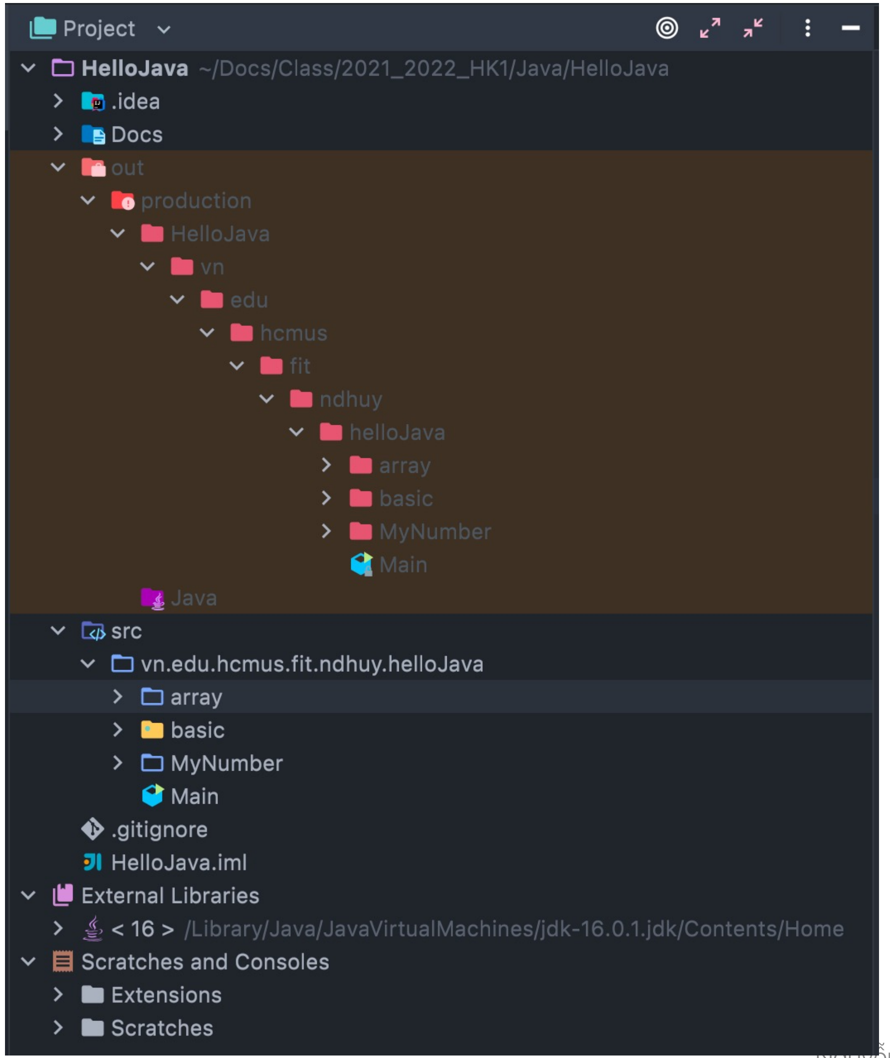
Project Structure

Project Structure



Project Structure

```java
package gui;
import java.util.Scanner;
import java.io.*;
/**
* @author NDHuy
*/
public class Main{
    /**
    * @param args the command line arguments
    */
    public static void main(String [] args){
          //TODO code application logic here
           . . .
    }
}
```

Comment

- Option 1:

```
// Comment on one line
```

- Option 2:

```
/*
 *    Comment with multiple-lines
 */
```

- Option 3:

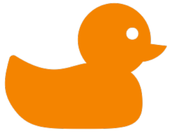
```
/* *
 *    Comment for Javadoc
 */
```

2

Data Type

Base data type

int, long



float, double

char, String

Date

Convert Number to String

```
String String.valueOf (Object obj);
String String.valueOf (boolean b);
String String.valueOf (char c);
String String.valueOf (char [] data);
String String.valueOf (char[] data, int offset, int count);
String String.valueOf (boolean b);
String String.valueOf (int i);
String String.valueOf (long l);
String String.valueOf (float f);
String String.valueOf (double d);
```

Convert to Integer

```
// int
Integer Integer.valueOf (String str);
int i = Integer.valueOf (String str).intValue();
Integer Integer.valueOf (int i);
int i = Integer.parseInt (String str);
// long
Long Long.valueOf (String str);
long i = Long.valueOf (String str).longValue();
Long Long.valueOf (long l);
Long l = Long.parseLong (String str);
```

Convert to Float

```
// float
Float Float.valueOf (String str);
float i = Float.valueOf (String str).floatValue();
Float Float.valueOf (int i);
float f = Float.parseFloat (String str);
// long
Double Double.valueOf (String str);
double i = Double.valueOf (String str).doubleValue();
Double Double.valueOf (double l);
double d = Double.parseDouble (String str);
```

Convert to Date

```
//day/month/year
SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy");
Date d= sdf.parse("2/2/2000");
//year-month-day
SimpleDateFormat sdf = new SimpleDateFormat("yyyy-MM-dd";
Date d= sdf.parse("2000-1-30");
//month/day/year
SimpleDateFormat sdf = new SimpleDateFormat("MM/dd/yyyy");
Date d= sdf.parse("3/20/2000");
```

Constant

- Define constant:

  final dataType CONSTANTNAME = Value;

- Example:

  final double PI = 3.14;

Operator

- Arithmetic operators: +, -, *, /, %
- Increment/Decrement operators: ++, --
- Assignment operators: +=, -=, *=, /= , %=
- Relational (Comparison) operators: <, <=, ==, !=, >=, >
- Conditional operators: &&, ||, !=, ==, &, | , !

3

Console input/output

Input from console

- Library:

```
java.util.Scanner;
```

- Uses:

```
Scanner scan = new Scanner (System.in);
String str   = scan.nextLine ();
```

Output to console

- Syntax:

```
System.out.print(…);
System.out.println(…);
```

- Example:

```
System.out.print("Hello Java");
System.out.println("Đại Học Khoa Học Tự Nhiên");
```

Format to print

- %b : boolean
- %c : char
- %d : int, long
- %f : float, double
- %s : String

Format

```
// Format
 String.format(“format”, parameter_1, …, parameter_n);
// Example
 int n = 100;
 double m = 20.8;
 String str = String.format (“n = %d, m= %f”, n, m);
 System.out.println (str);
```

```
String studentID=“0212055”;
String studentName = “Nguyễn Đức Huy”;
Double score = 9.5;
// String plus
System.out.println(studentID + “-” + studentName + “:” + score);
// Format to output
String str = String.format(“%s - %s : %f”, studentID, studentName, score);
System.out.println(str);
```

4

Basic Syntax

if

```
if (codition) {
    Line code 1;
    …
    Line code n;
}
```

```
Scanner scan = new Scanner (System.in);
String str   = scan.nextLine ();
int n = Integer.parseInt(str);
if ( n > 0 ) {
    System.out.println("n is positive number");
}
```

if - else

```
if (condition) {
    block of code if the condition is true;
} else {
    block of code if the condition is false;
}
```

```
System.out.println("n=");
Scanner scan = new Scanner(System.in);
int n = Integer.parseInt(scan.nextLine());
if (n > 0) {
            System.out.println("n is positive number");
} else {
      System.out.println("n is negative number");
}
```

if - else if - else

```
if (condition) {
        …
}else if (condition) {
        …
}
else if (condition) {
        …
}
else {
        …
}
```

if - else if - else

```
Scanner scan = new Scanner(System.in);
System.out.print(“Enter Grade Point Average:”);
int gpa = Integer.parseInt(scan.nextLine());
if (gpa >= 9) {
       System.out.println(“Outstanding”);
} else if (gpa >= 8 && gpa < 9) {
       System.out.println(“Excellent”);
}
else if (gpa >= 7 && gpa < 8) {
       System.out.println(“Good”);
}
else if (gpa >= 5 && gpa < 7) {
    System.out.println(“Average”);
}
else {
   System.out.println(“Below Average”);
}
```

switch - case

```
switch (value){
      case value_1:
            …
            break;
      …
      case value_N:
            …
            break;
      default:
            …
}
```

```
value : char, byte, short, int
value_1, value_2 have the same data-type with value
```

switch - case

```
int month = 8;
String str;
switch (month) {
    case 1:  str = "January";               break;
    case 2:  str = "February";              break;
    . . .
    case 11: str = "November";              break;
    case 12: str = "December";              break;
    default: str = "The month is invalid";  break;
}
System.out.println(str);
```

while

```
while (loop_codition) {
    …
}
```

```
Scanner scan = new Scanner (System.in);
System.out.print("Enter n:");
int n = Integer.parseInt(scan.nextLine());
int s=0;
int i=1;
while (i<=n) {
    s = s + I;
}
System.out.println("S="+s);
```

do while

```
do{
      ......
}while (loop_codition)
```

```
Scanner scan = new Scanner (System.in);
int n;
do{
    System.out.print("Enter n:");
    n = Integer.parseInt(scan.nextLine());
}while (n < 3 || n>=10)
```

for

```
for (initial variable; loop_codition; action after loop){
        …
}
```

```
Scanner scan = new Scanner (System.in);
System.out.print("Enter n:");
int n = Integer.parseInt(scan.nextLine());
int s=0;
for (int i=1; i<=n; i++){
    s = s + i;
}
System.out.println("S="+s);
```

break

- Scope
  - *for, while, do-while*
- Uses
  - Exit loop cycle

```
Scanner scan = new Scanner (System.in);
System.out.print("Enter n:");
int n = Integer.parseInt(scan.nextLine());
for (int i=1; i<=n; i++){
    System.out.print(i+" ");
    if(i==5){
        break;
    }
}
```

continue

- Scope
  - *for, while, do-while*
- Uses
  - Skip current step and go to next loop-cycle

```
Scanner scan = new Scanner (System.in);
System.out.print("Enter n:");
int n = Integer.parseInt(scan.nextLine());
for (int i=1; i<=n; i++){
    if(i==5){
        continue;
    }
    System.out.print(i+" ");
}
```

Math

```
double Math.abs (double d)
double Math.sqrt (double d)
double Math.min (double a, double b)
double Math.max (double a, double b)
double Math.exp (double e)
double Math.round (double r)
double Math.pow (double a, double b)
double Math.random ()
double Math.PI
double Math.E
```

Math

```java
//sqrt
double d = 9.5;
double s = Math.sqrt (d);
//pow
double a = 2.3;
double b = Math.pow (b, 3);
//min - max
double c = 6;
double d = 9.2;
double min = Math.min (c, d);
double max = Math.max (c, d);
```

5

References

References

- Books
  - Nguyễn Hoàng Anh, slide and video for Java programming course, FIT@HCMUS, 2010
  - The Java Language Specification Third Edition (2005)
- Online references
  - Java documentation: https://docs.oracle.com/en/java/


ngiyabonga
рахмат
danke
謝謝
spas
teşekkür ederim
Баярлалаа
mersi
kia ora
barka
welalin
tack
mahalo
tapadh leat
спасибо
faafetai lava
vinaka
blagodaram
dank je
misaotra
paldies
grazzi
matondo
спасибі
thank you
gracias
kiitos
dankie
nanni
nandri
dhanyavad
хвала
bedankt
mauruuru
asante
manana
obrigada
tenki
enkosi
bayarlalaa
gràçie
hvala
köszönöm
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dankon
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arigatô
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mèsi
didi madloba
ありがとう
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grazie
najis tuke
rahmet
kam sah hamnida
rahmat
terima kasih
diolch
dhanyavadagalu
shukriya
mercé
мерси
তোমাকে ধন্যবাদ
감사합니다
xièxie
ευχαριστώ
merci